THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** the sunlight leaf	Starch made.
2	**(b)** 80°	i + r = 40 + 40.
3	**(b)** have continuous daylight	No change in facing.
4	**(b)** transpiration	Water vapour condenses.
5	**(d)** 20 m	Ratio 2:1.
6	**(b)** evening sky, in the west	Soon after sunset.
7	**(a)** taking food from its host	Parasite.
8	**(c)** half	Inversely.
9	**(a)** umbra	Total.
10	**(a)** remain brown	No chlorophyll.
11	**(c)** 65°	90 - 25.
12	**(a)** no seasons	Same sunlight all year.
13	**(d)** guard cells	Guard cells swell.
14	**(c)** noon	Sun overhead.
15	**(b)** sunset	Opposite the Sun.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	More light for photosynthesis (1); Greater food production (1).	1+1
17	Reflection 35° (1); Between rays 70° (1).	1+1
18	Axis is tilted (1); Sun above horizon for different durations (1).	1+1
19	Less water is lost in sunlight (1); Gas exchange continues (1).	1+1
20	It reflects more light (1); Image is bright (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Plant in a closed jar with KOH (1); Control with water (1); Test leaves for starch: only control positive (1).	1+1+1
22	Incidence = 90 - 70 = 20° (1); Reflection 20° (1); Between rays 40° (1).	1+1+1
23	New moon aligned with Sun and Earth (1); Moon's umbra falls on Earth (1); Region in umbra sees total eclipse (1).	1+1+1
24	Guard cells swell and the pore opens (1); They shrink and it closes (1); Light and water matter (1).	1+1+1
	OR: Shows the road ahead (1); Helps avoid accidents (1); Wide view (1).	1+1+1
25	Near the poles (1); Sun above horizon at midnight (1); During summer (1).	1+1+1
	OR: Hydrilla under a funnel in water (1); Test tube over stem (1); Gas collects; glowing splinter relights (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Broad flat surface (1); Thin (1); Stomata for gases (1); Veins for transport (1); Chloroplasts with chlorophyll (1).	1+1+1+1+1
	OR: Normal drawn (1); Angle of incidence 30° (1); Angle of reflection 30° (1); Angle between incident and reflected 60° (1); Reflected ray on opposite side of the normal (1).	1+1+1+1+1
27	21 June: north summer (1); 22 Dec: north winter (1); Equinoxes: equal day and night (1); South opposite (1); Tilt same direction (1).	1+1+1+1+1
	OR: Water the plant (1); Cover branch with a clear bag (1); Seal the stem (1); Place in sunlight (1); Droplets inside show transpiration (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** dead wood	Saprotroph.
2	**(d)** away from the hole	Longer distance.
3	**(d)** penumbra	Partial shadow.
4	**(b)** remains brown	No light.
5	**(d)** 0°	Retraces its path.
6	**(a)** northern	North tilted to Sun.
7	**(c)** water and minerals	Large area.
8	**(c)** sunrise or sunset	Sun low.
9	**(c)** sunrise	Along with the Sun.
10	**(d)** trapping insects	Gets nitrogen.
11	**(d)** blurred and brighter	Wide beam.
12	**(c)** the Earth's shadow is large	Larger shadow.
13	**(d)** no starch is formed	No CO2.
14	**(a)** 2 m	1 m from the mirror, so 2 m.
15	**(a)** summer	Opposite.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	No air or water (1); No weathering (1).	1+1
17	Rhizobium fixes nitrogen (1); Soil fertility improves (1).	1+1
18	Two reflections (1); Image remains upright (1).	1+1
19	Rays slant and spread (1); Shorter days (1).	1+1
20	No photosynthesis (1); Stored starch is used up (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Regular reflection in mirror (1); Diffused in wall (1); Rough surface scatters (1).	1+1+1
22	Lower gravity (1); One-sixth (1); Easier (1).	1+1+1
23	Water goes up via xylem (1); Food moves via phloem (1); Direction differs (1).	1+1+1
24	Height 1.4 m (1); Image 3 m behind the mirror (1); Distance from her 6 m (1).	1+1+1
	OR: Lamp as Sun (1); Ball moved around (1); Lit part visible changes (1).	1+1+1
25	More light at noon (1); More photosynthesis (1); Less light at dawn (1).	1+1+1
	OR: Smooth surface gives regular reflection (1); Ripples scatter light (1); Image is distorted (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Solar: Moon between Sun and Earth (1); Lunar: Earth between (1); Umbra and penumbra (1); Straight travel of light (1); Occurs rarely (1).	1+1+1+1+1
	OR: The Moon turns once on its axis in about 27 days (1); It also goes round the Earth in about 27 days (1); So the same face points towards the Earth (1); The far side is never seen from the Earth (1); The far side still gets sunlight (1).	1+1+1+1+1
27	Potted plant kept in dark 48 h (1); Cover part of a leaf with black paper (1); Expose to sunlight (1); Test with iodine (1); Only uncovered part blue-black (1).	1+1+1+1+1

	OR: Closer object gives larger image (1); Farther object gives smaller (1); Larger screen distance gives larger image (1); Brightness decreases (1); Inverted always (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** cells lose water and become limp	Water loss.
2	**(a)** east	Sun in the west.
3	**(d)** half disc, right half lit (northern hemisphere)	Quarter of the way.
4	**(b)** makes food by photosynthesis	Provides food.
5	**(d)** the hole is covered	No light enters.
6	**(a)** dark reddish	Sunlight bends through atmosphere.
7	**(d)** more bubbles in sunlight	Photosynthesis.
8	**(a)** 1.2 m tall	Same size.
9	**(d)** 12 hours	Equal.
10	**(b)** reduce water loss	Conserve water.
11	**(d)** decreases	Fewer rays blocked.
12	**(b)** the Sun always shines on half of it	One side faces Sun.
13	**(d)** gives shelter and absorbs water	Protection.
14	**(b)** real image	Light actually reaches.
15	**(b)** lunar	Only lunar.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	New distance 1 m from mirror (1); Image 1 m behind: 2 m apart (1).	1+1
17	Sun's rays are always nearly direct (1); Day length is nearly constant (1).	1+1
18	It takes food from the host (1); It does not make food (1).	1+1
19	Compare shadows of tree and stick (1); Ratio gives the height (1).	1+1
20	Time between two full moons (1); About 29.5 days (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Kharif in rains (1); Rabi in winter (1); Temperature and water matter (1).	1+1+1
22	Rhizobium fixes nitrogen (1); Soil enriched (1); Saves cost (1).	1+1+1
23	Look above water (1); While submerged (1); Two mirrors (1).	1+1+1
24	The Moon blocks the Sun's light (1); Umbra falls on the place (1); Day looks like dusk (1).	1+1+1
	OR: Leaf forms pitcher (1); Slippery rim and lid (1); Digestive juices (1).	1+1+1
25	Box with a hole and screen (1); Candle in front (1); Inverted image on the screen (1).	1+1+1
	OR: June longer (1); December shorter (1); Axis tilt (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	New moon (1); Waxing crescent, first quarter (1); Waxing gibbous, full moon (1); Waning gibbous, last quarter (1); Waning crescent (1).	1+1+1+1+1
	OR: Lichen: alga and fungus (1); Alga food, fungus shelter (1); Rhizobium and legumes (1); Plant gives sugar (1); Bacteria fix nitrogen (1).	1+1+1+1+1
27	New distance from the mirror = 3 m (1); The image is 3 m behind the mirror (1); Distance between her and image = 6 m (1); Image distance equals object distance (1); The image is virtual (1).	1+1+1+1+1
	OR: Night light (1); Lunar calendar (1); Festivals (1); Moon missions (1); Phases teach geometry of light (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND